

STUDENT BOOK ANSWERS

Chapter 5 Calculating chemical quantities

Worked example 5.1

- a** $M_r = 31 \times 4 = 124$
- b** $M_r = 12 + 16 \times 2 = 44$
- c** $M_r = 1 + 14 + 16 \times 3 = 63$
- d** $M_r = 12 \times 6 + 1 \times 12 + 16 \times 6 = 180$

Worked example 5.2

- a** $M_r = 27 + 14 = 41$
- b** $M_r = 137.3 + 14 \times 2 + 16 \times 6 = 261.3$ (261)
- c** $M_r = 23 \times 2 + 12 + 16 \times 3 = 105.98$ (106)
- d** $M_r = 63.5 + 32.1 + 16 \times 4 = 159.6$ (160)

Question set 5.1

- 1**
- a** $M_r = 14 \times 2 = 28$
- b** $M_r = 1 \times 2 + 32.1 = 34.1$
- c** $M_r = 30.97 \times 2 + 16 \times 5 = 141.9$
- d** $M_r = 12 \times 12 + 1 \times 22 + 16 \times 11 = 342$
- e** $M_r = 1 \times 2 + 32.1 + 16 \times 4 = 98.1$
- 2**
- a** $M_r = 39.1 + 35.5 = 74.6$
- b** $M_r = 24.3 + 19 \times 2 = 62.3$
- c** $M_r = 40.1 \times 3 + 31 \times 2 + 16 \times 8 = 310.3$
- d** $M_r = 63.5 + 14 \times 2 + 16 \times 6 = 187.5$
- e** $M_r = 14 \times 2 + 1 \times 8 + 32.1 + 16 \times 4 = 132.1$
- 3** $(14 \times 2) = \text{N}_2$ or $(12 + 16) = \text{CO}$
- H_2O , CO_2 , NO_2 have, respectively, molecular masses of M_r : $2 + 16 = 18$, $12 + 16 \times 2 = 44$ and $14 + 2 \times 16 = 46$.

Worked example 5.3

$$\text{a } C = \frac{12}{12 + 4 \times 1} \times 100 = 75\%$$

$$H = \frac{4 \times 1}{12 + 4} \times 100 = 25\%$$

$$\text{b } Na = \frac{23}{23 + 35.5} \times 100 = 39\%$$

$$Cl = \frac{35.5}{23 + 35.5} \times 100 = 61\%$$

$$\text{c } Ca = \frac{40.1}{40.1 + 2(12 + 14)} \times 100 = \frac{40.1}{92.1} \times 100 = 43.5\%$$

$$C = \frac{2 \times 12}{92.1} \times 100 = 26\%$$

$$N = \frac{28}{92.1} \times 100 = 30.5\%$$

$$\text{d } N = \frac{14 \times 2}{14 \times 2 + 1 \times 8 + 12 \times 2 + 4 \times 16} \times 100 = \frac{28}{124} \times 100 = 22.6\%$$

$$H = \frac{8}{124} \times 100 = 6.5\%$$

$$C = \frac{2 \times 12}{124} \times 100 = 19.4\%$$

$$O = \frac{4 \times 16}{124} \times 100 = 51.6\%$$

Question set 5.2

- Percentage composition is the percentage by mass of each element.
- A chemist would want to know percentage composition when, for example, analysing an unknown compound to determine the amount of each element present.

$$\text{3 a } C = \frac{12}{12 + 16 + 2 \times 35.5} \times 100 = \frac{12}{99} \times 100 = 12.1\%$$

$$O = \frac{16}{99} \times 100 = 16.2\%$$

$$Cl = \frac{35.5 \times 2}{99} \times 100 = 71.7\%$$

$$\text{b } K = \frac{39.1 \times 2}{39.1 \times 2 + 32.1 + 4 \times 16} \times 100 = \frac{39.1 \times 2}{174.3} \times 100 = 44.9\%$$

$$S = \frac{32.1}{174.3} \times 100 = 18.4\%$$

$$O = \frac{16 \times 4}{174.3} \times 100 = 36.7\%$$

$$4 \quad \text{Cl} = \frac{3 \times 35.5}{12 \times 8 + 1 \times 5 + 3 \times 16 + 35.5 \times 3} \times 100 = 41.7\%$$

5 a Chalcocite (Cu_2S) has highest percentage Cu (79.9%)

b Chalcopyrite (CuFeS_2) has lowest percentage Cu (34.6%)

c Bornite (Cu_5FeS_4) would be the best to mine as it has the lowest % S of the two

Worked example 5.4

$$a \quad n(\text{C}) = \frac{N}{N_A} = \frac{3.3 \times 10^{25}}{6.02 \times 10^{23}} = 54.8 \text{ mol}$$

$$b \quad n(\text{H}_2\text{O}) = \frac{N}{N_A} = \frac{1.2 \times 10^{22}}{6.02 \times 10^{23}} = 0.02 \text{ mol}$$

$$c \quad n(\text{Na}) = \frac{N}{N_A} = \frac{6.6 \times 10^{21}}{6.02 \times 10^{23}} = 0.011 \text{ mol}$$

$$d \quad n(\text{CaCO}_3) = \frac{N}{N_A} = \frac{3.01 \times 10^{24}}{6.02 \times 10^{23}} = 5.0 \text{ mol}$$

$$e \quad n(\text{H}_2) = \frac{N}{N_A} = \frac{9.05 \times 10^{23}}{6.02 \times 10^{23}} = 1.5 \text{ mol}$$

Worked example 5.5

$$a \quad \text{Number molecules of CH}_4 = N = nN_A = 0.1 \times 6.02 \times 10^{23} = 6.02 \times 10^{22}$$

$$b \quad \text{Number of H atoms} = nN_A = 6.0 \times 6.02 \times 10^{23} = 3.61 \times 10^{24}$$

$$c \quad \text{Number of Al ions} = nN_A = 0.5 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$$

$$d \quad \text{Number of Ar atoms} = nN_A = 0.082 \times 6.02 \times 10^{23} = 4.94 \times 10^{22}$$

$$e \quad \text{Number of NH}_3 \text{ molecules} = nN_A = 2.1 \times 6.02 \times 10^{23} = 1.26 \times 10^{24}$$

Worked example 5.6

$$a \quad n(\text{K}^+) = 3 \text{ mol}, n(\text{Cl}^-) = 3 \text{ mol}$$

$$b \quad n(\text{H}) = 6 \times 12 = 72 \text{ mol}$$

$$c \quad n(\text{atoms}) = nN_A = 0.05 + 1.5 = 2 \text{ mol or } 4 \text{ atoms per molecule} \times 0.5 \text{ mol} = 2 \text{ mol of atoms}$$

$$d \quad \text{Number O atoms} = nN_A = 3 \times 2.5 \times 6.02 \times 10^{23} = 4.52 \times 10^{24}$$

$$e \quad \text{Number atoms} = nN_A = (2 + 5 + 1 + 1) \times 1.25 \times 6.02 \times 10^{23} = 6.77 \times 10^{24}$$

Question set 5.3

1 The Avogadro constant (N_A) is the number of atoms (6.02×10^{23}) in exactly 12 grams of the carbon – 12 isotope.

2 A mole is the amount of substance containing 6.02×10^{23} particles of that substance.

3 A mole is the amount of substance containing Avogadro's constant number of particles.

$$4 \text{ a } n(\text{He}) = \frac{N}{N_A} = \frac{4 \times 10^{24}}{6.02 \times 10^{23}} = 6.64 \text{ mol}$$

$$\text{b } n(\text{Cl}_2) = \frac{N}{N_A} = \frac{1.03 \times 10^{22}}{6.02 \times 10^{23}} = 0.017 \text{ mol}$$

$$\text{c } n(\text{Cu}) = \frac{N}{N_A} = \frac{1.204 \times 10^{23}}{6.02 \times 10^{23}} = 0.2 \text{ mol}$$

$$\text{d } n(\text{Al}_2\text{O}_3) = \frac{N}{N_A} = \frac{8.22 \times 10^{21}}{6.02 \times 10^{23}} = 0.0137 \text{ mol}$$

$$5 \text{ a } \text{Number of O}_2 \text{ molecules} = nN_A = 2.5 \times 6.02 \times 10^{23} = 1.505 \times 10^{24}$$

$$\text{b } \text{Number of Au atoms} = nN_A = 0.33 \times 6.02 \times 10^{23} = 1.99 \times 10^{23}$$

$$\text{c } \text{Number of Fe ions} = nN_A = 1.6 \times 6.02 \times 10^{23} = 9.63 \times 10^{23}$$

$$\text{d } \text{Number of AgNO}_3 \text{ units} = nN_A = 0.035 \times 6.02 \times 10^{23} = 2.11 \times 10^{22}$$

$$6 \text{ a } n(\text{Cu}^{2+}) = 3 \text{ mol}$$

$$\text{b } n(\text{CO}_3^{2-}) = 3 \text{ mol}$$

$$\text{c } n(\text{O}) = 9 \text{ mol}$$

$$\text{d } \text{Number O atoms} = nN_A = 3 \times 3 \times 6.02 \times 10^{23} = 5.42 \times 10^{24}$$

$$7 \text{ a } n(\text{C}_8\text{H}_{18}) = \frac{N}{N_A} = \frac{3 \times 10^{26}}{6.02 \times 10^{23}} = 498 \text{ mol}$$

$$\text{b } n(\text{C}) = 498 \times 8 \text{ mol} = 3987 \text{ mol}$$

$$\text{c } n(\text{H}) = 498 \times 18 \text{ mol} = 8964 \text{ mol}$$

$$8 \text{ a } n(\text{O}) = 2.5 \times 4 = 10 \text{ mol}$$

$$\text{b } n(\text{O}) = 0.2 \times 2 = 0.4 \text{ mol}$$

$$\text{c } n(\text{Al}_2(\text{CO}_3)_3) = \frac{N}{N_A} = \frac{3.02 \times 10^{24}}{6.02 \times 10^{23}} = 5.02; n(\text{O}) = 9 \times n(\text{Al}_2(\text{CO}_3)_3) = 9 \times 5.02 = 45.1 \text{ mol}$$

Worked example 5.7

$$\text{a } M_r(\text{Br}_2) = 79.9 \times 2 = 159.8 \text{ g}$$

$$\text{b } M_r(\text{H}_2\text{SO}_4) = (1 \times 2 + 32.1 + 16 \times 4) = 98.1 \text{ g}$$

$$\text{c } M_r(\text{Ca}(\text{OH})_2) = 40.1 + 16 \times 2 + 1 \times 2 = 74.1 \text{ g}$$

$$\text{d } M_r(\text{NH}_4\text{NO}_3) = 14 \times 2 + 1 \times 4 + 16 \times 3 = 80 \text{ g}$$

$$\text{e } M_r(\text{CH}_3\text{CH}_2\text{OH}) = 12 \times 2 + 1 \times 6 + 16 = 46 \text{ g}$$

Worked example 5.8

$$\text{a } n(\text{Zn}) = \frac{m}{M_r} = \frac{13}{65.4} = 0.199 \text{ mol}$$

$$\text{b } n(\text{Cl}_2) = \frac{m}{M_r} = \frac{7.5}{35.5 \times 2} = 0.106 \text{ mol}$$

$$\text{c } n(\text{PbI}_2) = \frac{m}{M_r} = \frac{3.35}{207.2 + 126.9 \times 2} = 0.0073 \text{ mol}$$

$$\text{d } n(\text{Cu}) = \frac{m}{M_r} = \frac{3.5 \times 1000}{63.5} = 55.1 \text{ mol}$$

Worked example 5.9

$$\text{a } m(\text{Fe}) = nM_r = 4.2 \times 55.8 = 234.6 \text{ g}$$

$$\text{b } m(\text{CO}) = nM_r = 0.045 \times (12 + 16) = 1.26 \text{ g}$$

$$\text{c } m(\text{Na}_2\text{CO}_3) = nM_r = 0.25 \times (23 \times 2 + 12 + 16 \times 3) = 26.5 \text{ g}$$

$$\text{d } m(\text{NH}_3) = nM_r = 1.34 \times (14 + 1 \times 3) = 22.8 \text{ g}$$

Question set 5.4

1 Relative molecular is the mass of one molecule or formula unit of a substance measured in amu while molar mass is the mass of one mole or 6.02×10^{23} particles of the substance measured in grams.

$$\text{2 a } M_r = 2 \times 2 + 2 \times 16 = 34 \text{ g}$$

$$\text{b } M_r = 24.3 + 2 \times 79.9 = 184.1 \text{ g}$$

$$\text{c } M_r = 40.1 + 12 + 3 \times 16 = 100.1 \text{ g}$$

$$\text{d } M_r = 12 \times 5 + 1 \times 12 = 72 \text{ g}$$

$$\text{3 a } m = nM_r = 7.1 \times 10^{-3} \times 20.2 = 0.14 \text{ g}$$

$$\text{b } m = nM_r = 5.43 \times 207.2 = 1125 \text{ g}$$

$$\text{c } m = nM_r = 3.14 \times 10^{-4} \times 4 \times 31 = 0.039 \text{ g}$$

$$\text{d } m = nM_r = 1.236 \times (24.3 + 32.1 + 4 \times 16) = 148.8 \text{ g}$$

$$\text{e } m = nM_r = 0.452 \times (3 \times 39.1 + 31 + 4 \times 16) = 96.0 \text{ g}$$

$$\text{4 a } n = \frac{m}{M_r} = \frac{10.00}{18} = 0.56 \text{ mol}$$

$$\text{b } n = \frac{m}{M_r} = \frac{10.00}{60} = 0.17 \text{ mol}$$

$$\text{c } n = \frac{m}{M_r} = \frac{10.00}{2 \times 12 + 6 \times 1} = 0.33 \text{ mol}$$

$$\text{d } n = \frac{m}{M_r} = \frac{10.00}{23 + 79.9} = 0.097 \text{ mol}$$

$$\text{e } n = \frac{m}{M_r} = \frac{10.00}{137.3 + 2 \times 35.5} = 0.048 \text{ mol}$$

$$5 \text{ a } n = \frac{m}{M_r} = \frac{0.028}{12 + 2 \times 16} = 0.00064 \text{ mol or } 6.4 \times 10^{-4} \text{ mol}$$

$$\text{b } m = nM_r = 3.0 \times 44 = 132 \text{ g}$$

$$6 \text{ a } m = nM = 1.00 \times (14 \times 12 + 18 + 2 \times 14 + 5 \times 16) = 1.00 \times 294 = 294 \text{ g}$$

$$\text{b } n = \frac{m}{M} = \frac{6.22}{294} = 0.021 \text{ mol}$$

$$\text{c } m = nM = 0.245 \times 294 = 72.0 \text{ g}$$

$$\text{d } N = nN_A = \frac{4.28 \times 10^{-3}}{294} \times 6.02 \times 10^{23} = 8.76 \times 10^{18} \text{ molecules}$$

$$7 \text{ a } n = \frac{m}{M} = \frac{4.00}{2 \times 16} = 0.125 \text{ mol ; } N = nN_A = 0.125 \times 6.02 \times 10^{23} = 7.53 \times 10^{22} \text{ molecules}$$

$$\text{b } 2 \times 7.53 \times 10^{22} \text{ molecules} = 1.506 \times 10^{23} \text{ atoms}$$

$$8 \text{ a } n = \frac{m}{M} = 80 \times 1000 \times \frac{0.8}{18} = 3556 \text{ mol}$$

$$\text{b } N = nN_A = 3556 \times 6.02 \times 10^{23} = 2.14 \times 10^{27} \text{ molecules}$$

$$9 \text{ } n = \frac{N}{N_A} = \frac{1}{6.02 \times 10^{23}} = 1.66 \times 10^{-24}$$

$$M = \frac{m}{n} = \frac{2.20 \times 10^{-22}}{1.66 \times 10^{-24}} = 132 \text{ g}$$

$$10 \text{ } M(\text{He}) = 4; M(\text{H}_2) = 2.0$$

The mixture has $N(\text{He}) = 2 \times N(\text{H}_2)$ and so, $n(\text{He}) = 2 \times n(\text{H}_2)$

$$m = nM \text{ and } m(\text{H}_2) + m(\text{He}) = 6 \text{ g}$$

$$\Rightarrow 6 = n(\text{H}_2) M(\text{H}_2) + n(\text{He}) M(\text{He}) = n(\text{H}_2) \times 2 + n(\text{He}) \times 4 = 2n(\text{H}_2) + 2n(\text{H}_2) \times 4 = 10n(\text{H}_2) = 6$$

$$n(\text{H}_2) = \frac{6}{10} = 0.6 \text{ mol; } m(\text{H}_2) = nM = 0.6 \times 2 = 1.2 \text{ g}$$

Worked example 5.10

a i	C	Br
mass	7.00	93.00
mol	$\frac{7.00}{12}$	$\frac{93.00}{79.9}$
ratio	0.583	1.16
	$\frac{0.583}{0.583}$	$\frac{1.16}{0.583}$
	1	2
	CBr ₂	

ii	C	H	O
mass	64.9	13.5	21.6
mol	$\frac{64.9}{12}$	$\frac{13.5}{1.0}$	$\frac{21.6}{16}$
ratio	5.41	13.5	1.35
	$\frac{5.41}{1.35}$	$\frac{13.5}{1.35}$	$\frac{1.35}{1.35}$
	4.0	10	1



iii	C	H	O	N
mass	49.5	5.2	16.6	28.8
mol	$\frac{49.5}{12}$	$\frac{5.2}{1.0}$	$\frac{16.6}{16}$	$\frac{28.8}{14}$
ratio	4.125	5.2	1.04	2.06
	$\frac{4.125}{1.04}$	$\frac{5.2}{1.04}$	$\frac{1.04}{1.04}$	$\frac{2.06}{1.04}$
	3.966	5	1	1.98



b $\text{C}_4\text{H}_{10}\text{O}$ EF mass = $4 \times 12 + 10 + 16 = 74$

$$\frac{\text{Relative molecular mass}}{\text{relative empirical mass}} = \frac{371}{74} = 5.01$$

Molecular formula = $5 \times \text{Empirical formula} = \text{C}_{20}\text{H}_{50}\text{O}_5$

Question set 5.5

1 Empirical formula is the simplest whole number ratio of the elements in a compound while molecular formula is the actual number of atoms of each element in a molecule.

2 a COCl_2

b C_2H_5

c NO_2

d Na_2PO_3

3	Cl	O
mass	100 – 18.4	18.4
mol	$\frac{81.6}{35.5}$	$\frac{18.4}{16}$
ratio	2.299	1.15
	$\frac{2.299}{1.15}$	$\frac{1.15}{1.15}$
	2	1



4 a	C	H	N
mass	74.0%	8.7	17.3
mol	$\frac{74}{12}$	$\frac{8.7}{1}$	$\frac{17.3}{14}$
ratio	6.17	8.7	1.24
	6.17/1.24	8.7/1.24	1.24/1.24
	5	7	1



b Molar mass = 162 EF mass = 81 so ratio 2 and molecular formula = $\text{C}_{10}\text{H}_{14}\text{N}_2$

5	Pb	O
mass	20.7	23.9 – 20.7
mol	$\frac{207}{207.2}$	$\frac{3.2}{16}$
ratio	0.1	0.2
	1	2



6 a $n(\text{CO}_2) = \frac{m}{M} = \frac{10.38}{44} = 0.236 \text{ mol CO}_2$

$$n(\text{H}_2\text{O}) = \frac{m}{M} = \frac{5.66}{18} = 0.314 \text{ mol H}_2\text{O}$$

$$(0.236 \text{ mol CO}_2, 0.314 \text{ mol H}_2\text{O})$$

b $n(\text{CO}_2) = n(\text{C}) = 0.236 \text{ mol C}$, $n(\text{H}) = 2n(\text{H}_2) = 0.628 \text{ mol H}$

c $m = nM = 0.236 \times 12 = 2.83 \text{ g C}$

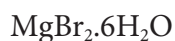
$$m = nM = 0.628 \times 1 \text{ g} = 0.628 \text{ g H}$$

d $m(\text{O}) = 7.24 - 2.83 - 0.628 = 3.782 \text{ g}$

e	C	H	O
mass	2.83	0.628	3.782
mol	$\frac{2.83}{12}$	$\frac{0.628}{1}$	$\frac{3.782}{16}$
ratio	0.235	0.628	0.236
	$\frac{0.235}{0.235}$	$\frac{0.628}{0.235}$	$\frac{0.236}{0.235}$
	1	2.7	1



7	MgBr ₂	H ₂ O
mass	4.60	(7.3 – 4.6)
mol	$\frac{4.6}{24.3 + 79.9 \times 2}$	$\frac{2.7}{18}$
ratio	0.025	0.15
	$\frac{0.025}{0.025}$	$\frac{0.15}{0.025}$
	1	6



- 8 Assuming the students were trying to determine the formula of an *oxide of copper*, and the following results were obtained:

Mass of dry crucible = 6.10 g

Mass of crucible + metal oxide = 14.95 g

Mass of crucible + metal = 13.95 g

a $m(\text{Cu}) = 13.95 - 6.10 = 7.85 \text{ g}$

b $m(\text{O}) = 13.60 - 12.60 = 1.00 \text{ g}$

c	Cu	O
mass	7.85	1.00
mol	$\frac{7.85}{63.6}$	$\frac{1.00}{16.0}$
	0.123	0.0625
	$\frac{0.123}{0.0625}$	$\frac{0.0625}{0.0625}$
	1.97	1

Cu₂O is the empirical formula

- d** To prevent the metal reacting with the air and reforming the oxide

Worked example 5.11

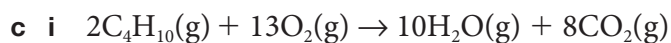
a i $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{MgO}(\text{s})$ $n(\text{O}) = \frac{1}{2}n(\text{Mg}) = 0.5 \times 4.5 = 2.25 \text{ mol}$

ii $n(\text{Mg}) = n(\text{MgO}) = 4.25 \text{ mol}$

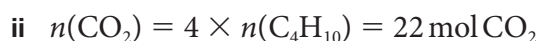
b i $n(\text{Fe}) = \frac{1}{2}n(\text{HCl}) = 0.5 \times 1.5 = 0.75 \text{ mol}$

ii $n(\text{H}_2) = \frac{1}{2}n(\text{HCl}) = \frac{1}{2} \times 3 = 1.5 \text{ mol}$

iii $n(\text{Fe}) = n(\text{FeCl}_2) = 4 \text{ mol}$



$$n(\text{O}_2) = \frac{13}{2} n(\text{C}_4\text{H}_{10}) = \frac{13}{2} \times 5.5 = 35.75 \text{ mol}$$



$$n(\text{H}_2\text{O}) = 5 \times n(\text{C}_4\text{H}_{10}) = 5 \times 5.5 = 27.5 \text{ mol H}_2\text{O}$$

Worked example 5.12

a i $n(\text{CaCO}_3) = \frac{m}{M} = \frac{25}{100.1} = 0.25$

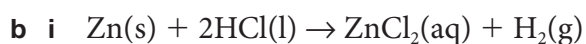
$$n(\text{CaCO}_3) = n(\text{CaO}) = 0.25$$

$$m = nM_r = 0.25 \times 56.1 = 14 \text{ g}$$

ii $n(\text{CaCO}_3) = \frac{m}{M} = \frac{25}{100.1} = 0.25$

$$n(\text{CO}_2) = n(\text{CaCO}_3) = 0.25$$

$$m = nM_r = 0.25 \times 44 = 11 \text{ g}$$



$$n(\text{Zn}) = \frac{m}{M} = \frac{2.6}{65.4} = 0.0398$$

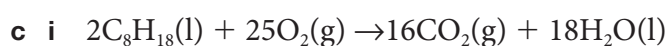
$$n(\text{ZnCl}_2) = n(\text{Zn}) = 0.0398$$

$$m = nM_r = 0.0398 \times (65.4 + 2 \times 35.5) = 5.4 \text{ g}$$

ii $n(\text{Zn}) = \frac{m}{M} = \frac{2.6}{65.4} = 0.0398$

$$n(\text{Zn}) = n(\text{H}_2) = 0.0398$$

$$m = nM_r = 0.0398 \times 2 = 0.080 \text{ g}$$



$$n(\text{octane}) = \frac{m}{M} = \frac{45.6}{114} = 0.4$$

$$n(\text{O}_2) = 12.5 n(\text{octane}) = 5 \text{ mol}$$

$$m = nM_r = 5 \times 32 = 160 \text{ g}$$

ii $n(\text{octane}) = \frac{m}{M} = \frac{45.6}{114} = 0.4$

$$n(\text{CO}_2) = 8 n(\text{octane}) = 3.2 \text{ mol}$$

$$m = nM_r = 3.2 \times 44 = 141 \text{ g}$$



$$n(\text{glucose}) = \frac{m}{M} = \frac{45}{6 \times 12 + 12 + 6 \times 16} = 0.25$$

$$n(\text{C}_2\text{H}_5\text{OH}) = 2 \times n(\text{glucose}) = 0.5 \text{ mol}$$

$$m = nM_r = 0.5 \times 46 = 23 \text{ g}$$

Question set 5.6

- 1 Stoichiometry is the study of the amounts of reactants and products in a chemical reaction.
- 2 The Law of conservation of mass states that in a chemical reaction mass is neither created nor destroyed.
- 3 A balanced chemical equation gives the mole ratio which is also the ratio of molecules.

4

Fe_2O_3	CO	Fe	CO_2
1 mol	3	2	3
1.5	4.5	3 mol	4.5
0.2	0.6 mol	0.4	0.6
0.03	0.09	0.06	0.09 mol

- 5 The total mass of products and total mass of reactants are equal.



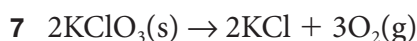
Ratio is 3:2; three mole of hydrogen will make 2 mole of ammonia, so 20 moles of ammonia requires

$$30 \text{ mol of hydrogen: } n(\text{H}_2) = \frac{3}{2} n(\text{NH}_3) = \frac{3}{2} \times 20 = 30 \text{ mol}$$

$$\text{b } n(\text{H}_2) = \frac{m}{M} = \frac{3.41}{2} = 1.705 \text{ mol}$$

$$n(\text{NH}_3) = \frac{2}{3} n(\text{H}_2) = \frac{2}{3} \times 1.705 = 1.14 \text{ mol}$$

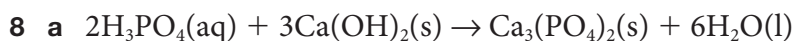
$$m = nM = 1.14 \times 17 = 19.4 \text{ g}$$



$$n(\text{O}_2) = \frac{m}{M} = \frac{80.5}{32} = 2.52 \text{ mol}$$

$$n(\text{KCl}) = \frac{2}{3} n(\text{O}_2) = \frac{2}{3} \times 2.52 = 1.68 \text{ mol}$$

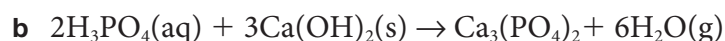
$$m = nM = 1.68 \times (39.1 + 35.5) = 125 \text{ g}$$



$$n(\text{H}_3\text{PO}_4) = \frac{m}{M} = \frac{30.0}{98} = 0.306 \text{ mol}$$

$$n(\text{Ca}(\text{OH})_2) = \frac{3}{2} n(\text{H}_3\text{PO}_4) = \frac{3}{2} \times 0.306 = 0.459 \text{ mol}$$

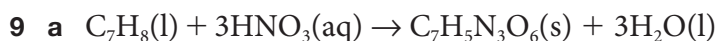
$$m = nM = 0.459 \times (40.1 + 2(16 + 1)) = 34 \text{ g}$$



$$n(\text{H}_3\text{PO}_4) = \frac{m}{M} = \frac{30.0}{98} = 0.306 \text{ mol}$$

$$n(\text{Ca}_3(\text{PO}_4)_2) = \frac{1}{2} n(\text{H}_3\text{PO}_4) = \frac{1}{2} \times 0.306 = 0.153 \text{ mol}$$

$$m = nM = 0.153 \times (40.1 \times 3 + 32 \times 2 + 16 \times 8) = 47.8 \text{ g}$$



$$n(\text{C}_7\text{H}_5\text{N}_3\text{O}_6) = \frac{m}{M} = \frac{1000000}{7 \times 12 + 5 + 3 \times 14 + 6 \times 16} = \frac{1000000}{227} = 4405 \text{ mol}$$

$$n(\text{C}_7\text{H}_8) = n(\text{C}_7\text{H}_5\text{N}_3\text{O}_6) = 4405 \text{ mol}$$

$$m = nM = 4405 \times (7 \times 12 + 8) = 405260 \text{ g} = 405 \text{ kg}$$

b $n(\text{HNO}_3) = 3 \times n(\text{C}_7\text{H}_5\text{N}_3\text{O}_6) = 3 \times 4405 = 13215 \text{ mol}$

$$m = nM = 13215 \times (1 + 14 + 3 \times 16) = 832545 \text{ g} = 833 \text{ kg}$$



The molar mass of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} = (2 \times 23 + 12 + 3 \times 16 + 10 \times 18) = 286 \text{ g mol}^{-1}$

The molar mass of $\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O} = (2 \times 23 + 12 + 3 \times 16 + 18) = 124 \text{ g mol}^{-1}$ or $286 - (9 \times 18)$

10 kg of washing soda: $n(\text{washing soda}) = \frac{m}{M} = \frac{10000}{286} = 34.9 \text{ mol}$

$$n(\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}) = n(\text{washing soda}) = 34.9 \text{ mol}$$

$$m(\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}) = nM = 34.9 \times 124 = 4.34 \text{ kg}$$

Each kg of powder cost $\frac{\$30}{4.34} = \6.91 .

The retailer makes a loss for each bag.

Chapter review questions

- Relative atomic mass is the mass of the atom compared to the mass of a carbon-12 atom being 12 exactly. Relative molecular mass is the mass of one molecule compared to the mass of a carbon-12 atom. It is determined by adding up the relative atomic masses as given by the molecular formula.
- Relative molecular mass is the mass of one molecule compared to one twelfth of a carbon 12 atom. Relative formula mass is used for ionic substances, it is the mass of one unit of the salt as shown by the formula.
- Percentage compositions allows chemists to determine the formula for the substances. It helps to determine the ratio of the elements which can then be used to determine the empirical formula and then the molecular formula. Knowledge of the formula of substances allows chemist to balance chemical equations, to control chemical reactions and to predict how much reactant is needed or product will be made.
- Avogadro's constant is the number of particles in one mole of the substance. By definition one mole of any substance contains the same number of particles as there are atoms in 12 g of carbon, or 6.02×10^{23} .
- Relative molecular mass is the mass of one molecule of the substance compared to one twelfth of a carbon 12 atom. The molar mass is the mass of one mole of the substance or (6.02×10^{23}) particles of that substance).
 - Empirical formula is the simplest whole number ratio of the elements present. Molecular formula is the actual ratio of the elements present.
 - Yes. The empirical formula for an ionic compound is the same as the formula as ionic substances have an infinite lattice structure.
- Law of conservation of mass.

$$8 \text{ a } M_r = 32 + 2 \times 16 = 64 \text{ g mol}^{-1}$$

$$\text{b } M_r = 39.1 + 14 + 3 \times 16 = 101.1 \text{ g mol}^{-1}$$

$$\text{c } M_r = 65.4 + 2 \times (16 + 1) = 99.4 \text{ g mol}^{-1}$$

$$\text{d } M_r = 2 \times 12 + 2 \times 16 + 4 = 60 \text{ g mol}^{-1}$$

$$9 \text{ a } \% \text{ Ca} = \frac{40.1}{40.1 + 2 \times 17} \times 100 = 54.1\%$$

$$\text{b } \% \text{ O} = \frac{16 \times 9}{16 \times 9 + 3 \times 12 + 2 \times 27} \times 100 = 61.5\%$$

$$10 \text{ a } \text{Ni: } \frac{58.7}{58.7 + 32.1 + 16 \times 4} \times 100 = 38\%$$

$$\text{S: } \frac{32.1}{58.7 + 32.1 + 16 \times 4} \times 100 = 20.7\%$$

$$\text{O: } \frac{16 \times 4}{58.7 + 32.1 + 16 \times 4} \times 100 = 41.3\%$$

$$\text{b } \text{H: } \frac{3 \times 1}{3 + 31 + 4 \times 16} \times 100 = 3\%$$

$$\text{P: } \frac{31}{3 + 31 + 4 \times 16} \times 100 = 32\%$$

$$\text{O: } \frac{16 \times 4}{3 + 31 + 4 \times 16} \times 100 = 65\%$$

$$11 \text{ a } n = \frac{N}{N_A} = \frac{3.01 \times 10^{24}}{6.02 \times 10^{23}} = 5 \text{ mol}$$

$$\text{b } n = \frac{N}{N_A} = \frac{1.05 \times 10^{22}}{6.02 \times 10^{23}} = 0.17 \text{ mol}$$

$$\text{c } n = \frac{N}{N_A} = \frac{1.204 \times 10^{28}}{(6.02 \times 10^{23})} = 2 \times 10^4 \text{ mol}$$

$$12 \text{ a } N = nN_A = 0.30 \times 6.02 \times 10^{23} = 1.8 \times 10^{23}$$

$$\text{b } N = nN_A = 4.0 \times 6.02 \times 10^{23} = 2.4 \times 10^{24}$$

$$\text{c } N = nN_A = 7.5 \times 6.02 \times 10^{23} = 4.5 \times 10^{24}$$

13 Molecular mass refers to the actual number of atoms of each element in a molecule. Ionic substances are not molecules. They have an infinite lattice structure; the numbers would be too large.

$$14 \text{ a } n = \frac{m}{M} = \frac{1.01}{20.2} = 0.05 \text{ mol}$$

$$\text{b } n = \frac{m}{M} = \frac{2.538}{126.9 \times 2} = 0.01 \text{ mol}$$

$$\text{c } n = \frac{m}{M} = \frac{5.1}{27 \times 2 + 3 \times 16} = 0.05 \text{ mol}$$

15 a $m = nM = 2.05 \times 85.5 = 175.3 \text{ g}$

b $m = nM = 1.5 \times (2 + 32.1 + 4 \times 16) = 1.5 \times 98.1 = 147 \text{ g}$

c $m = nM = 1.0 \times (6 \times 12 + 12 + 6 \times 16) = 1.0 \times 180 = 180 \text{ g}$

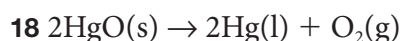
d $m = nM = 0.1 \times (23.0 + 35.5) = 5.85 \text{ g}$

16 The formula gives the chemist the simplest ratio of the elements in the molecule.

17 a $n(\text{CO}_2) = \frac{4}{2} n(\text{C}_2\text{H}_6) = 2 \text{ mol}$

b $n(\text{H}_2\text{O}) = \frac{6}{2} n(\text{C}_2\text{H}_6) = 3 \times 3 = 9 \text{ mol}$

c $n(\text{O}_2) = \frac{7}{2} n(\text{C}_2\text{H}_6) = \frac{7}{2} \times 0.6 = 2.1 \text{ mol}$



$n(\text{O}_2) = \frac{1}{2} n(\text{HgO}) = \frac{1}{2} \times 0.24 = 0.12 \text{ mol}$

19 a $n(\text{H}) = 4 n(\text{urea}) = 8 \text{ mol}$

b $n(\text{O}) = n(\text{urea}) = 2 \text{ mol}$

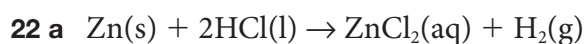
c $n(\text{N}) = 4 \text{ moles}, N = nN_A = 4 \times 6.02 \times 10^{23} = 2.4 \times 10^{24} \text{ particles}$

20	Cu	O
mass	38.205 – 35.030	39.005 – 38.205
	3.175 g	0.800 g
mol	$\frac{3.175}{63.5}$	$\frac{0.800}{16}$
ratio	0.050 mol	0.050 mol
	$\frac{0.050}{0.050}$	$\frac{0.050}{0.050}$
	1	1

CuO

21	Mg	Cl
mass	0.98	3.8 – 0.98
	0.98	2.8
mol	$\frac{0.98}{24.3}$	$\frac{2.8}{35.5}$
ratio	0.04	0.79
	$\frac{0.04}{0.04}$	$\frac{0.0790}{0.04}$
	1	2

This supports an empirical formula of MgCl_2 for this compound.



$$n(\text{Zn}) = n(\text{ZnCl}_2) = 0.012 \text{ mol}$$

$$m = nM = 0.012 \times 65.4 = 0.785 \text{ g}$$

b $n(\text{ZnCl}_2) = \frac{1}{2} n(\text{HCl}) = \frac{1}{2} \times 0.040 = 0.020 \text{ mol}$

$$m = nM = 0.020 \times 136.4 = 2.73 \text{ g}$$

23 Assume mass of bones is 11 kg, 50% is calcium phosphate.

$$n(\text{calcium phosphate}) = \frac{11000 \times 0.5}{3 \times 40.1 + 2 \times 31 + 8 \times 16} = 17.7 \text{ mol}$$

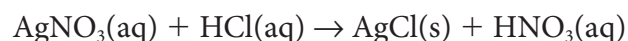
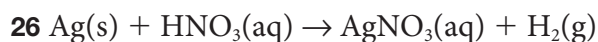
$$n(\text{P}) = 2n(\text{calcium phosphate}) = 35.4 \text{ mol}$$

$$m = nM = 35.4 \times 31.0 = 1099 = 1.1 \text{ kg}$$

24 $n(\text{H}_2) = \frac{m}{M} = \frac{1}{2} = 0.5 \text{ mol}$. $n(\text{H atoms}) = 2n(\text{H}_2) = 1.0 \text{ mol}$. A sample with the same number of atoms would have the same number of moles so $n(\text{Zn}) = 1.0 \text{ mol}$. $m = nM = 1.0 \times 65.4 = 65.4 \text{ g}$

25 a	Fe 20.09%	S 11.55%	H 5.04%	O 63.32%
	<u>20.09</u>	<u>11.55</u>	<u>5.04</u>	<u>63.32</u>
	55.8	32.1	1.0	16
	0.36	0.36	5.04	3.96
	<u>0.36</u>	<u>0.36</u>	<u>5.04</u>	<u>3.96</u>
	.36	0.36	0.36	0.36
	1	1	14	11
	$\text{FeSO}_{11}\text{H}_{14}$			

b In one formula unit there would be 7 water molecules and this would mean 7 oxygen molecules are also in the water molecules. The formula would then be $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$.



$$n(\text{AgCl}) = \frac{1.050}{107.9 + 35.5} = 0.00732 \text{ mol}$$

$$m = nM = 0.00732 \times 107.9 = 0.790 \text{ g}$$

$$\text{Percentage of silver} = \frac{0.790}{1.580} \times 100 = 50.0\%$$



$$n(\text{iron oxide}) = \frac{5.00}{55.8 \times 2 + 3 \times 16} = 0.0313 \text{ mol}$$

$$n(\text{Fe}) = 2n(\text{iron oxide}) = 2 \times 0.0313 \text{ mol}, m = nM = 0.0627 \times 55.8 = 3.496 \text{ g}$$

$$\text{Purity} = \frac{3.496}{3.58} \times 100 = 97.7\% \text{ pure}$$

$$\text{Or \% of Fe in iron oxide is } \frac{2 \times 55.8}{2 \times 55.8 + 3 \times 16} \times 100 = 69.9\%$$

$$69.9\% \text{ of } 5.0 \text{ g} = 3.496 \text{ g. Purity is } \frac{3.496}{3.58} \times 100 = 97.7\%$$